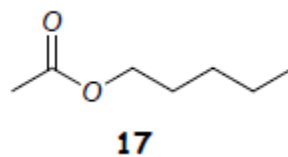
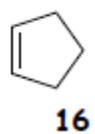
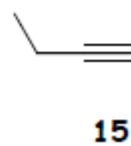
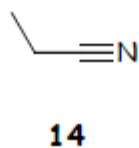
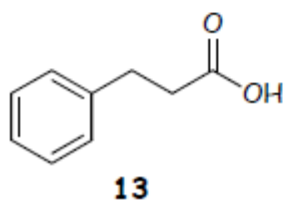
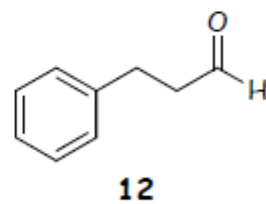
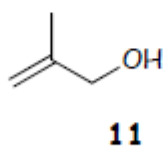
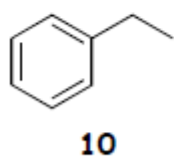
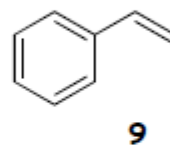
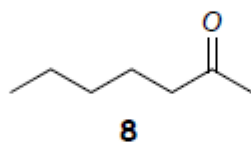
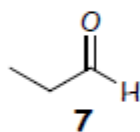
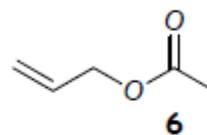
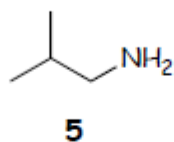
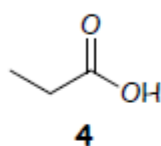
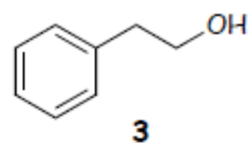
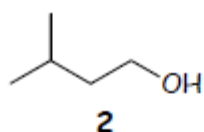
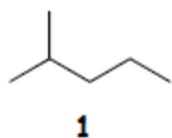


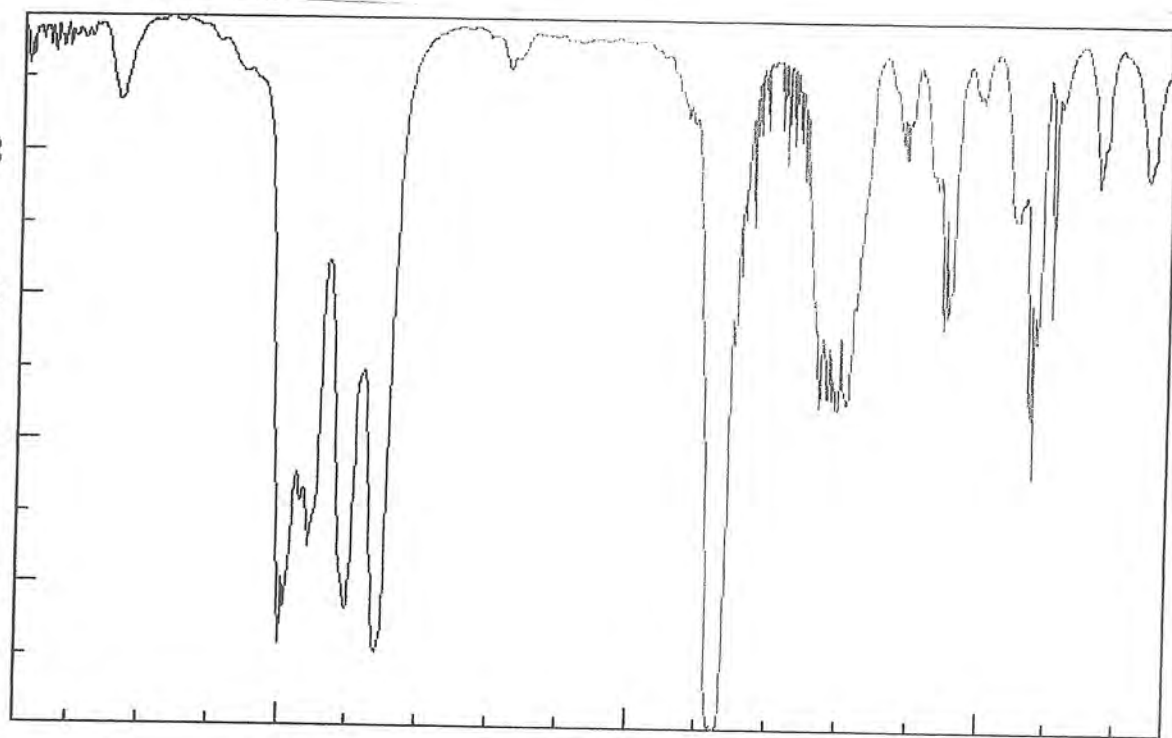
Q1. The IR spectra (A – F) of the six compounds are provided on the following pages. Each of the spectra is produced by one of 17 compounds that are shown below. Match each spectrum with the correct compound.



(A)

TRANSMITTANCE

0.8
0.6
0.4
0.2



3000

2000

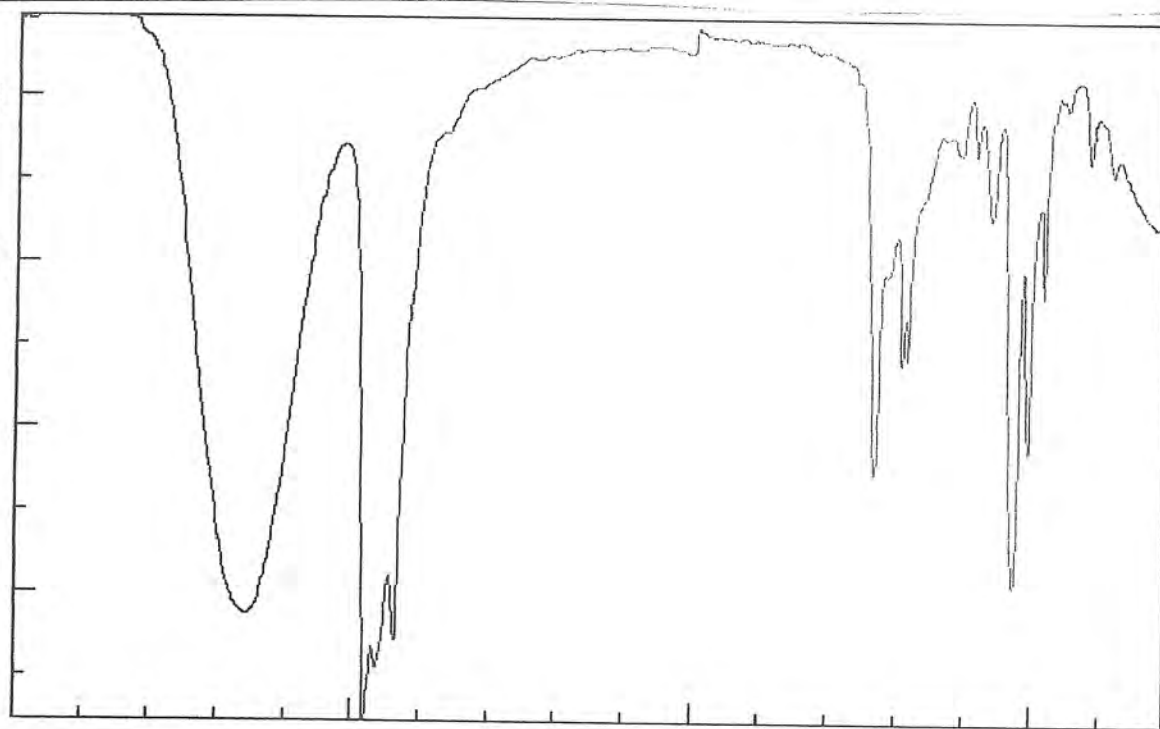
1000

Wavenumber (cm-1)

(B)

TRANSMITTANCE

0.8
0.6
0.4
0.2



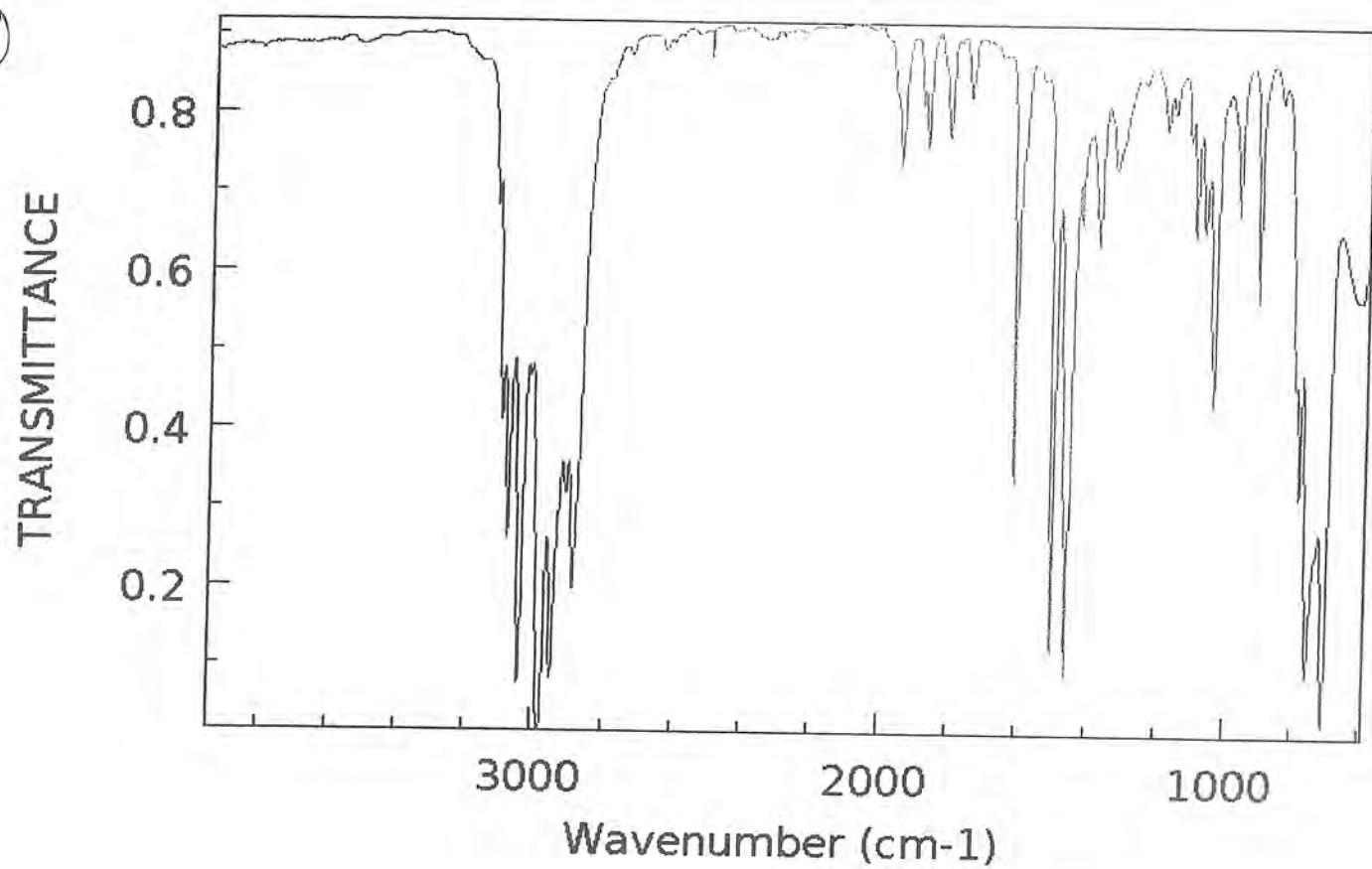
3000

2000

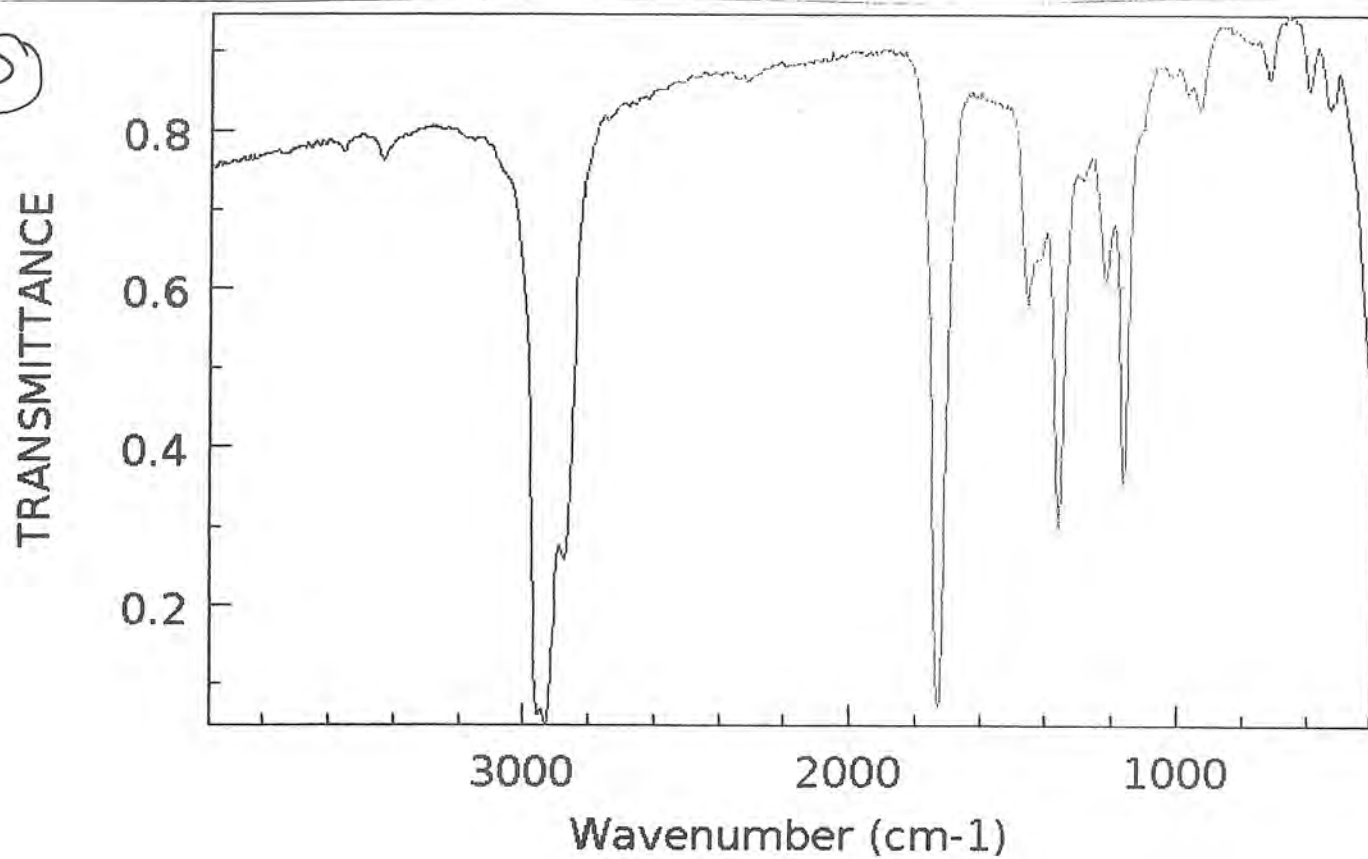
1000

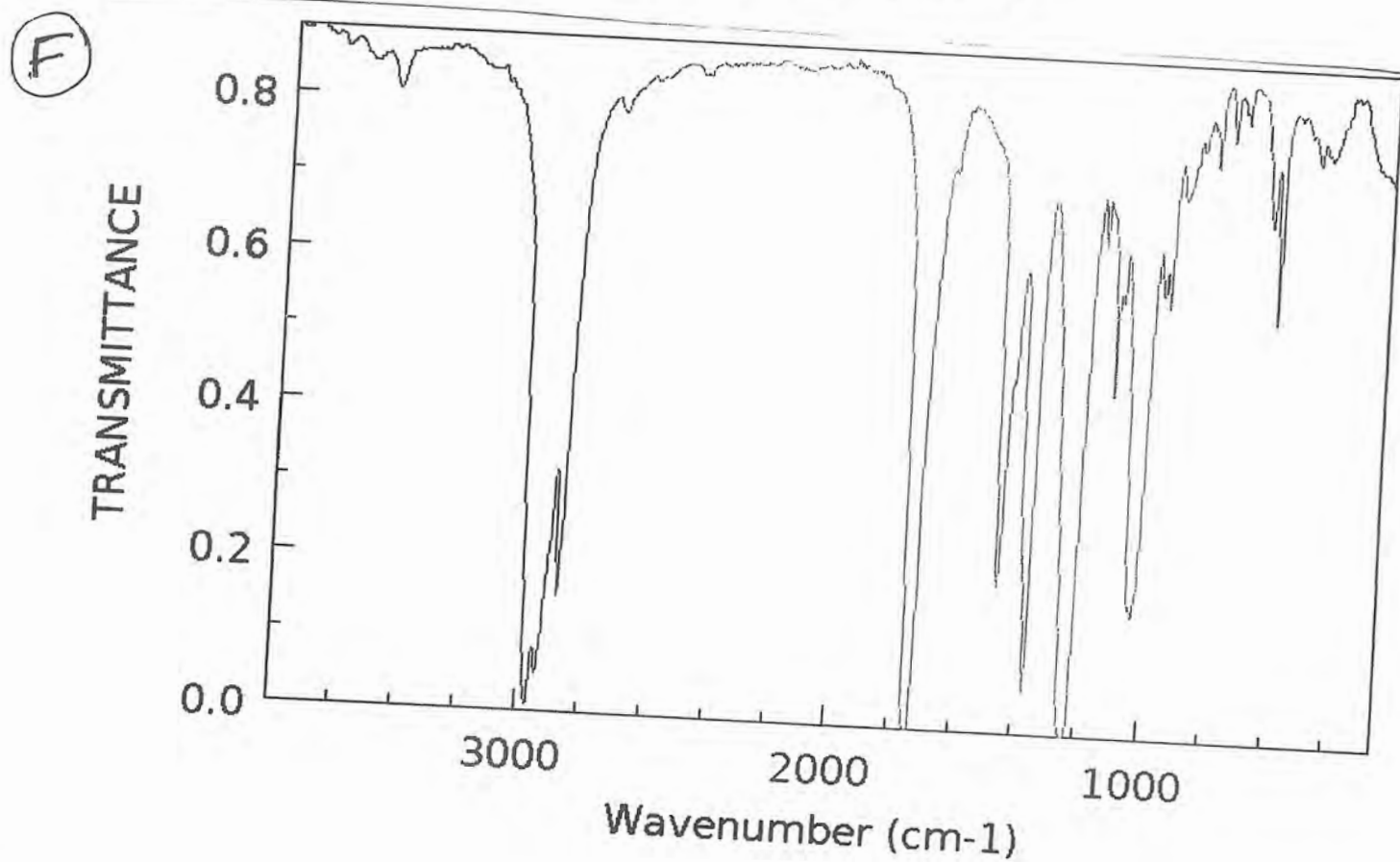
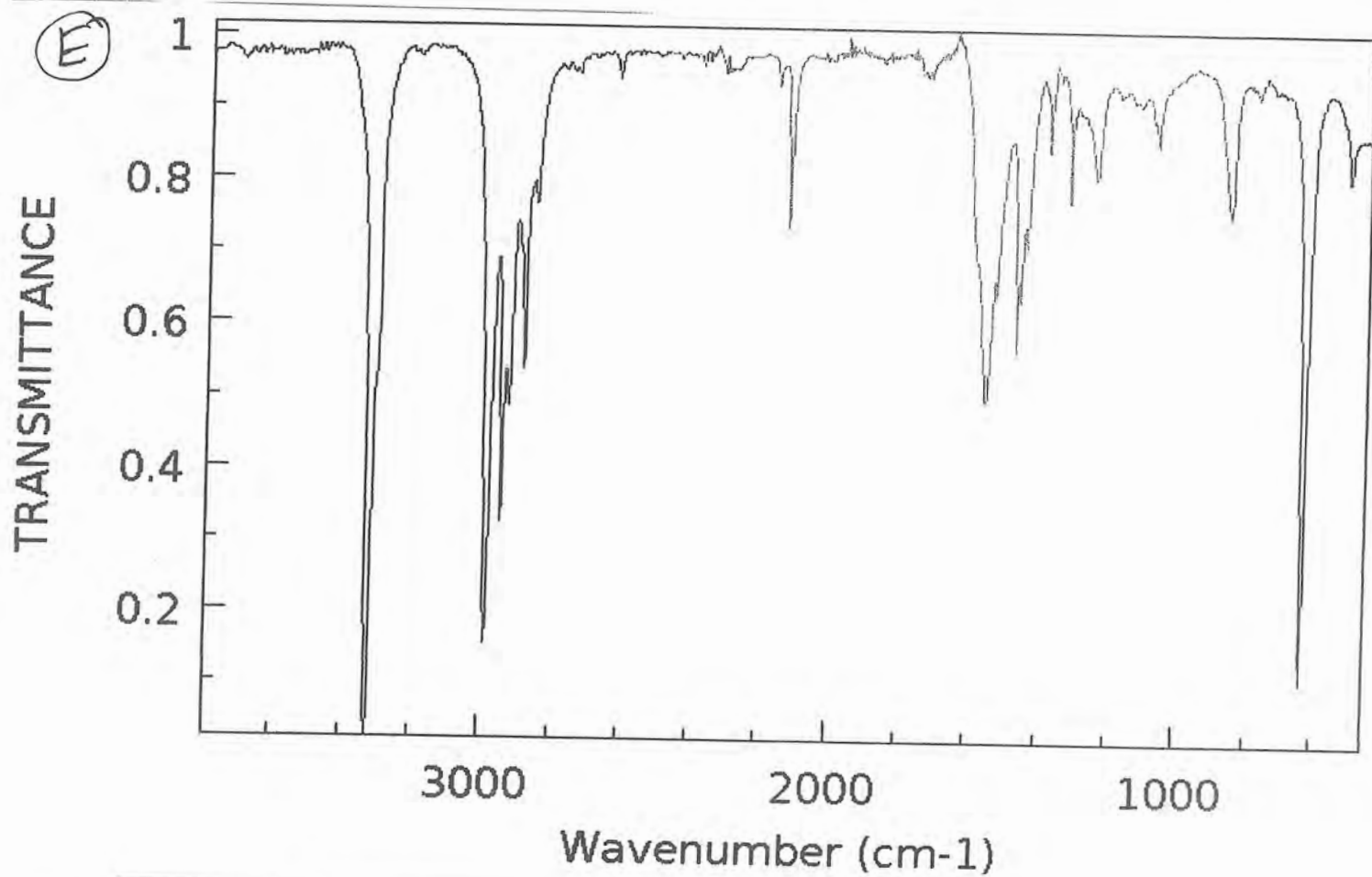
Wavenumber (cm-1)

(C)



(D)





Q2. You find a bottle on the shelf only labeled C_3H_6O . You take an IR spectrum of the compound and find major peaks at 2950, 1720, and 1400 cm^{-1} . Draw a molecule that might be the compound in the bottle.

Q3. You have an unknown that you are told is one of the four compounds below, and it exhibits the following spectral data. Which compound do you have?

IR major peaks: 1710 cm^{-1} Large and Sharp, 3300 cm^{-1} Large and Broad, 3000 cm^{-1} C-H Aliphatic (mostly obscured by 3300 cm^{-1} broad peak).

